

Australian IT Labour Market Intelligence Platform

Master Project Workflow & Development Roadmap

This document defines the development lifecycle of the project. The objective is to build a professional end-to-end analytics platform following the same workflow used in real-world data projects.

Project Pipeline

Research Design → Data Collection → Data Profiling → Data Cleaning → Data Standardisation → PostgreSQL → SQL Analytics → Tableau → Business Insights → Documentation

Phase	Objective
1. Research Design	Define business problem, research questions, project architecture, README
2. Data Collection	Collect raw data from APIs and official sources. Store immutable JSON/CSV
3. Data Profiling	Inspect structure, identify missing values, duplicates, data types and quality
4. Data Cleaning	Manual review in Excel followed by Python automation for large-scale cleaning
5. Data Standardisation	Merge all cleaned datasets into one consistent analytical schema.
6. PostgreSQL	Design relational schema, load curated data, create indexes and analytical
7. SQL Analytics	Perform exploratory analysis and answer business questions using SQL.
8. Tableau	Develop interactive dashboards for stakeholders.
9. Business Insights	Interpret findings and provide evidence-based recommendations.
10. Documentation	Complete README, Data Dictionary, Cleaning Report, SQL documentation a

Core Principle

Understand the data → Clean the data → Standardise the data → Model the data → Analyse the data → Visualise the results → Deliver business insights.

Project Status

Stage	Status
Research Design	Completed
Data Collection	In Progress
Data Profiling	Next
Remaining Phases	Planned